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Decoupling Structure, Style and Colour: A Modular Diffusion Pipeline for Photograph-to-Pixel-Art Conversion under Few-Shot Constraints

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A dissertation presented in part fulfilment of the requirements of the Degree of Master of Science at The University of Glasgow

9 September 2026

Abstract

This dissertation investigates the automated conversion of photographs into character portraits matching a specific pixel art aesthetic under severe constraints: a thirty-image dataset and a single consumer-grade graphics processing unit. The challenge lies in the representational gap between the continuous gradients of diffusion models and the discrete grid and limited palette of pixel art. A specific failure mode in few-shot style adaptation is documented: the acquisition of the target style occupies the adapter's limited capacity, overwriting the base model's attribute-binding capability and causing prompt-based colour control to fail. In response, a decoupled pipeline is proposed that assigns structural constraints to an edge-conditioned control network, stylistic character to a low-rank adapter, and colour to image-to-image conditioning. Quantitative evaluation reveals that fidelity and colour-statistic metrics rank the same outputs differently from a perceptual measure based on CLIP feature distance, demonstrating that an evaluation metric must be sensitive to the spatial organisation that defines the target domain. The text-only configuration falls inside the feature-space range spanned by the reference set itself, indicating that style acquisition is not the limiting factor; identity preservation is. Finally, the evaluated configurations form a Pareto frontier representing a structural trade-off between style and identity, on which the ideal region is empty. Rather than offering a strictly optimal solution, the complete pipeline occupies the middle of this frontier, providing the only configuration that is not extreme on either measure.

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Chapter 1: Introduction

1.1 Motivation

Pixel art holds a unique position in modern media. Initially a result of early hardware limits, it is now a deliberate artistic style driven by a large indie game industry, with major hits like Stardew Valley. However, creating pixel art is still highly manual. An artist must draw portraits pixel by pixel, often spending hours on a single small image. Diffusion models offer a natural way to automate this process. Following the success of latent diffusion [1], high-resolution image generation is now widely accessible, supported by a strong ecosystem of tools built around Stable Diffusion. It is therefore worth exploring whether a small studio or an individual could use these tools to automatically convert a real photograph into a pixel art portrait that fits a specific game. The main challenge is that diffusion models and pixel art work in fundamentally different ways. A model trained on photographs learns to produce smooth colour gradients and fine textures. Pixel art demands the exact opposite: a strict grid of square pixels, a limited palette of perhaps 16 to 32 colours, sharp boundaries, and the deliberate removal of complex details. The gap between them is not simply about resolution, but about the core rules of how an image is represented.

1.2 Aims and Objectives

The primary aim of this project is to determine whether an accurate and controllable photograph-to-pixel-art pipeline can be built under severe data and computing constraints. It also seeks to identify which parts of this process require machine learning and which are better handled by standard, deterministic algorithms. Four specific objectives support this aim. The first is to build a highly precise dataset from official game assets and fine-tune a style adapter with it. The second is to deeply analyse rather than just observe the failures that happen when this adapter generates images. The third is to design a preprocessing chain that converts any input photograph into a standard format, preparing it for structural control. The fourth is to compare different strategies for colour control, evaluating which method best balances preserving the subject's identity with matching the target style.

1.3 Contributions

This dissertation makes four main contributions. First, it diagnoses a specific failure in few-shot style adaptation: learning a strong new style overwrites the base model's ability to link colours to specific objects, causing it to completely ignore colour instructions in text prompts. This is shown to be a consistent result of training on a very small dataset for a highly abstract target style, rather than a random error. Second, it proposes and tests a modular architecture. Instead of relying on text to control everything, the pipeline separates the process so that structure, style, and colour are handled by three distinct mechanisms. Third, it directly compares two colour-control strategies under identical conditions and identifies the specific settings that yield the best results for the pipeline. Fourth, it provides a direct comparison of evaluation metrics, demonstrating that a colour-statistic measure ranks the four outputs differently at every position from a perceptual measure based on CLIP feature distance. This establishes that an evaluation metric must be sensitive to the specific property defining the target domain. For pixel art, that property is spatial organisation rather than global colour proportion.

1.4 Dissertation Structure

Section 2 reviews the relevant literature across four areas: diffusion-based synthesis, parameter-efficient adaptation, structural conditioning, and algorithmic pixel art. Section 3 describes the construction of the training dataset and the captioning problems encountered. Section 4 presents the four-stage pipeline architecture. Section 5 covers implementation, including the dependency and memory constraints imposed by the target environment. Section 6 reports experimental results, ablations and the quantitative baseline comparison. Section 7 discusses the findings and the limitations of the approach, and Section 8 concludes.

Chapter 2: Background and Related Work

2.1 Diffusion Models for Image Synthesis

Denoising diffusion probabilistic models generate images by learning to reverse a fixed forward process that progressively corrupts data with Gaussian noise [2]. Training reduces to a regression objective in which a network predicts the noise added at a randomly sampled timestep. Later work showed that the sampling trajectory need not follow the forward process step for step, reducing inference cost by an order of magnitude [3]. Diffusion models have since been shown to surpass generative adversarial networks on standard image synthesis benchmarks [4], establishing the paradigm as the dominant approach to image generation. The computational cost of operating directly in pixel space remained prohibitive until the diffusion process was moved into the latent space of a pretrained variational autoencoder [1], itself descended from the auto-encoding variational Bayes formulation [5]. This latent diffusion architecture compresses a 512 by 512 image into a 64 by 64 latent representation before denoising, reducing the spatial dimensionality by a factor of sixty-four. Text conditioning is supplied by a contrastive language-image encoder [6] and injected into the denoising network through cross-attention layers [7]. The denoising network itself follows an encoder-decoder topology with skip connections [8]. Classifier-free guidance [9] amplifies the influence of the conditioning signal at inference time by extrapolating between conditional and unconditional predictions. This project uses the publicly released Stable Diffusion 1.5 checkpoint. Later checkpoints [10] offer higher fidelity but require substantially more memory, and the ecosystem of control networks and adapters is more mature for the earlier version.

2.2 Parameter-Efficient Adaptation and Style Transfer

Neural style transfer originally treats style as a set of texture statistics to be matched by optimisation [11]. This works well for painterly textures but fails for pixel art, where the defining properties are a strict geometric grid and a limited colour palette, rather than texture. Paired and unpaired image translation frameworks [12], [13] offer an alternative, but both require much larger training datasets than are available here. Within the diffusion paradigm, three personalisation methods are relevant. Textual inversion [15] learns a new token embedding while freezing the model; it is extremely lightweight but struggles to represent a complete artistic style. DreamBooth [16] fine-tunes the entire network, achieving high fidelity but demanding significant memory and often overfitting on small datasets. Low-rank adaptation [17] occupies a middle ground: it freezes the original weights and learns a low-rank residual for selected projection matrices. This greatly reduces the number of trainable parameters compared to full fine-tuning.

2.3 Structural Conditioning of Diffusion Models

Text alone is a weak specification of spatial layout. Control networks address this by copying the diffusion model's encoder, feeding it an auxiliary image such as an edge map, and injecting the extracted features back into the frozen original network [19]. Because they start with zero-initialised weights, training remains stable even on small datasets. The conditioning image used in this project is produced by a standard Canny edge detector [21].

While learned edge detectors [22] produce perceptually cleaner contours, the available control network checkpoints for Stable Diffusion 1.5 are trained specifically on Canny outputs. Substituting a different detector would introduce a mismatch between training and inference. A second way to condition the model focuses on overall composition and colour rather than structure. By adding noise to a guide image and denoising from that point—rather than starting from pure noise—the output retains the guide's global colours while adopting the target style [23]. This allows colour information to be supplied as an image rather than text. This project adopts this approach, and Section 6.5 reports the comparison that motivated this choice.

2.4 Algorithmic Pixel Art Generation

Pixel art generation has a research history predating diffusion models. Early methods approached it as an optimisation problem, iteratively grouping pixels into superpixels and reducing the colour palette until the image converged [25], [26]. While these methods produce structurally sound pixel art, they are purely reductive: they cannot hallucinate stylistic features that are absent from the input, such as the specific facial proportions of a game's art direction. Section 6.7 uses one of these traditional methods as a baseline. Other approaches use deep learning, such as training networks to pixelise and depixelise images jointly without needing paired training data [28]. However, these are generally trained on broad datasets and are not tied to one specific artistic style. The present work differs from both traditions. Instead of treating pixelisation as a single learned or optimised transformation, it breaks the task apart. Reductive tasks (like scaling and colour limitation) are handled by standard deterministic algorithms [25], while generative tasks (like stylistic detailing) are left to the conditioned diffusion model. Section 7.1 argues that this decomposition is exactly why the pipeline succeeds where prompt-driven generation alone fails.

2.5 Colour Quantisation

Reducing a photograph to a small palette is a classical problem in computer graphics, historically handled by median cut [30] or octree [31] algorithms. Today, clustering approaches derived from k-means [32], [33] are the most widely used, with mini-batch variants [34] making them practical for large images. A common issue with basic clustering is that colours are prioritised by pixel count. In a portrait, hair and clothing take up most of the image, so perceptually important but spatially small colours like lip or eye tones are often swallowed by dominant clusters. Section 4.3 describes a weighting scheme introduced to counteract this. Additionally, edge-preserving smoothing is applied before quantisation using a bilateral filter [35]. This averages nearby pixels of similar colours, thereby suppressing texture without blurring the boundaries between distinct regions.

2.6 Research Gap

The literature reviewed above establishes that few-shot style adaptation is feasible, that structural conditioning is highly effective for edge maps, and that palette reduction has reliable algorithmic solutions. What is not established is how these components interact when the target domain is as abstract as pixel art, and the training data is as scarce as thirty images. In particular, while the literature on low-rank adaptation reports successful style acquisition and subject fidelity, it rarely examines what capabilities the base model loses in the process. This dissertation addresses that gap directly.

Chapter 3: Dataset Construction

3.1 Source Material and Selection Criteria

Thirty official character portraits from Stardew Valley were selected as the training corpus. Selection was governed by three criteria: the portrait must be a frontal or near-frontal bust composition, the subject must be a single character, and the image must be free of environmental background. These criteria match the desired output. Since the training budget is limited to thirty images, any variation in the dataset that is not needed in the final output simply wastes model capacity.



Figure 3.1: A representative training pair — source portrait and its manually written caption.

3.2 Lossless Upscaling and Background Normalisation

The native resolution of the source assets is substantially below the 512 by 512 input expected by the base model. Bilinear or bicubic upscaling introduces interpolated intermediate values along every pixel boundary, which destroys precisely the hard aliased edges that characterise the target style. Nearest-neighbour interpolation was therefore used throughout, replicating each source pixel into an integer-sized block and preserving both edge sharpness and the exact palette of the original asset. The source assets carry an alpha channel. Compositing them onto a pure white background rather than retaining transparency serves two purposes: it removes the risk that the model learns spurious structure from the alpha channel, and it fixes a consistent background convention that can later be requested at inference time.

3.3 Domain Hallucination in Automated Captioning

Automated captioning was attempted first, using a vision-language tagger of the type commonly applied to illustration datasets. The output was unusable. Because such taggers are trained predominantly on high-resolution anime and illustration corpora, they exhibited systematic domain hallucination when presented with pixel art: shading blocks were interpreted as discrete physical objects, dark pixel clusters were labelled as differently coloured hair, and in several cases the tagger failed to detect a human figure at all and emitted a no-person tag. This observation is worth stating precisely because it generalises. The failure is not one

of tagger quality but of representational mismatch: at the effective resolution of a pixel art portrait, the visual features on which such models depend simply do not exist. Any project attempting automated annotation of low-resolution stylised assets should expect the same behaviour.

3.4 Manual Semantic Disentanglement

Automated captioning was consequently abandoned in favour of manual annotation. Each of the thirty images received a comma-separated caption beginning with a fixed trigger token, followed by the character name, a count and framing descriptor, and then a minimal set of observable physical attributes: hair colour and length, facial hair, garment type and colour, expression, and background. The design principle was semantic disentanglement. The trigger token is intended to carry the style, and every other token is intended to carry a specific, independently variable attribute. If a style descriptor such as a palette characteristic were folded into the trigger token's context, the model would be unable to separate style from attribute at inference time. The extent to which this design succeeded is examined in Section 6.3, where it is shown that disentanglement at the caption level does not guarantee disentanglement in the learned representation.

Chapter 4: Methodology

4.1 Architecture Overview

The pipeline comprises four stages. Stage one, executed once, fine-tunes a low-rank style adapter on the dataset described in Section 3. Stage two converts an arbitrary input photograph into two derived representations: a quantised colour image and a binary edge map. Stage three performs conditioned generation, using the edge map for structure, the adapter for style, and the quantised image for colour. Stage four applies deterministic pixelisation to the generated result. The rationale for this division is that the three properties the output must satisfy have different natures. Structural correctness is a geometric constraint that can be specified exactly and enforced by conditioning. Stylistic character is a distributional property that must be learned. Colour fidelity is a mapping that can be computed in closed form. Handling these properties separately ensures no single mechanism is forced to perform tasks it is poorly suited for. The experimental record in Section 6 supports this division.

4.2 Stage One: Low-Rank Style Adaptation

The adapter is injected into the query, key, value and output projections of every attention block in the denoising network, and additionally into the input and output projections of each transformer block. Rank and scaling factor are both set to sixteen, giving a unit effective scaling. Dropout of 0.05 is applied to the adapter branch as regularisation against the small dataset. The text encoder and the variational autoencoder are frozen throughout. Only the adapter residuals receive gradients, amounting to approximately half of one per cent of the denoising network's parameters. Training minimises the mean squared error between predicted and actual noise at uniformly sampled timesteps, following the standard denoising objective [2]. Optimisation uses decoupled weight decay [36], with a cosine-annealed learning rate schedule [37] preceded by a linear warm-up of fifty steps. Two training configurations were run. The first used a learning rate of one times ten to the minus four for 450 steps at batch size two. The second reduced the learning rate to five times ten to the minus five and extended training to approximately 900 steps. Section 6.2 reports the difference between them.

4.3 Stage Two: Photograph Preprocessing

Input photographs vary in aspect ratio, background and lighting, none of which the downstream model is equipped to normalise. Preprocessing therefore proceeds in five deterministic steps.

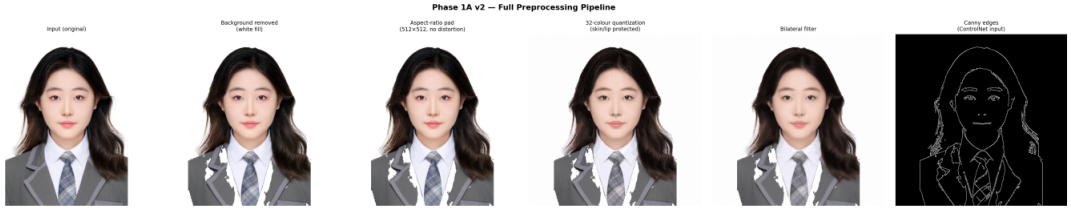


Figure 4.1: The five-stage preprocessing chain, from input photograph to Canny edge map.

Background removal is performed by semantic segmentation using a transformer-based network [38] fine-tuned for clothing and person parsing. The predicted person mask is used to composite the subject onto a white field, matching the convention established in the training set. Dedicated salient-object networks [39] are an alternative, and Section 7.3 notes a failure case that motivates revisiting this choice. Geometric normalisation preserves aspect ratio. The image is scaled so that its longer edge measures 512 pixels and the remainder of the square canvas is filled with white. Cropping was rejected because it removes clothing detail that carries identity information, and anisotropic scaling was rejected because it distorts facial proportions that the structural conditioning stage will subsequently enforce. Colour quantisation reduces the image to thirty-two colours by mini-batch clustering [34]. Standard clustering often loses important but rare colours, such as lip tones. To prevent this, pixels matching basic RGB skin and lip filters are oversampled by a factor of sixteen before clustering, and the resulting palette is then applied to the original image. These filters are deliberately broad: the cost of over-inclusion is a slightly larger share of palette capacity allocated to warm tones, whereas the cost of under-inclusion is the loss of the feature entirely. With oversampling disabled, the mean absolute deviation between original and quantised lip colour was 23.3 intensity levels; with the factor set to sixteen, this fell to 14.0. Edge-preserving smoothing is then applied using a bilateral filter [35], suppressing residual texture within quantised regions without softening the boundaries between them. Finally, a Canny detector [21] extracts the structural contour. Three threshold pairs were compared and the intermediate setting retained; Section 6.4 reports this comparison.

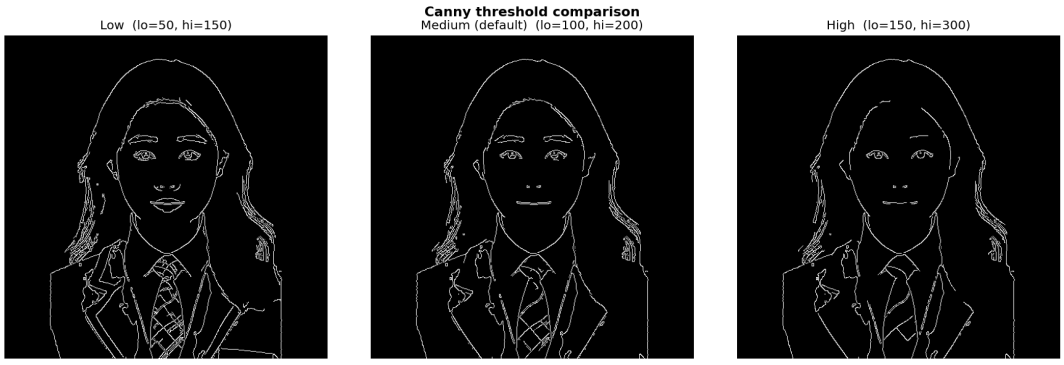


Figure 4.2: Canny threshold comparison at low, intermediate and high hysteresis settings.

4.4 Stage Three: Conditioned Generation

The generation stage receives three conditioning signals. The edge map is supplied to a control network [19], whose influence is modulated by a conditioning scale. The style adapter is merged into the denoising network weights before sampling. The quantised colour image is supplied as the initial state of the sampling trajectory [23], with a noise strength parameter controlling how far the sampler may depart from it. Sampling uses thirty steps and a classifier-free guidance scale of 7.5 throughout. The text prompt is deliberately minimal, consisting of the trigger token and a small number of compositional descriptors. Section 6.3 documents the empirical finding that motivates this restraint.

4.5 Stage Four: Deterministic Pixelisation

The generated image is downsampled by nearest-neighbour interpolation to one eighth of its resolution and immediately upsampled by the same method, producing uniform square cells aligned to a regular grid. This enforces the grid regularity that the adapter approximates but does not guarantee. An optional final step remaps every pixel to its nearest neighbour in the

palette extracted during stage two; Section 6.5 reports why this step is retained only as an alternative rather than as the default.

Chapter 5: Implementation

5.1 Environment and Dependency Resolution

The pipeline was implemented in Python and executed on a hosted notebook platform providing a single graphics processing unit with sixteen gigabytes of memory. This environment introduced several engineering constraints. They are documented here because they shaped the implementation and will likely affect anyone attempting to reproduce the work. Three distinct dependency conflicts were encountered. The diffusion library version required by the adapter implementation depends on a repository-access function that had been removed from the current release of the model hub client, causing an import failure at initialisation. This was resolved by pinning the hub client to a historical release before installing the diffusion library, and by suppressing transitive dependency resolution during that installation so that the package manager could not silently restore the incompatible version. The second conflict concerned the numerical computation library. The platform’s default numerical library was binary-incompatible with the required tensor library. Pinning the numerical library to a compatible release resolved this, but introduced a third conflict: the vision utility library is itself compiled against the newer numerical library and fails to import once the downgrade is applied. Rather than attempt to satisfy both, the image preprocessing was reimplemented using only the imaging library and array primitives, removing the dependency entirely.

5.2 Memory and Concurrency Constraints

Half-precision mixed training [40] was enabled to fit the denoising network, the autoencoder and the text encoder in memory simultaneously, and gradient checkpointing [41] was used to trade recomputation for activation storage. These memory-saving choices created a minor conflict: the mixed-precision scaler rejects half-precision gradients from the adapter. Casting only the adapter’s trainable weights to single precision immediately after injection resolved this while leaving the frozen majority of the network in half precision, so the memory saving is preserved. Data loading was restricted to a single process. Multi-process loading deadlocks silently in this environment, because the tokeniser maintains internal thread state that does not survive process forking. With a thirty-image dataset the throughput cost is negligible.

5.3 Adapter Weight Merging at Inference

The adapter must be composed with the base weights before inference. The library version in use does not expose an adapter-loading method on the denoising network object, so merging was implemented directly: for each adapted layer, the product of the two learned matrices is scaled and added to the frozen weight. Two details required care. Because the adapted layers include both 2D linear projections and 4D convolutions, the merge is handled uniformly by flattening the weights to two dimensions, computing their product, and reshaping them back to the original format. All tensors are held on the host during merging to avoid device mismatch, and the merged network is cast back to half precision before transfer to the accelerator.

Chapter 6: Experiments and Results

6.1 Experimental Setup

All experiments use the same base checkpoint, the same edge-conditioned control network, and a fixed random seed, so that differences between conditions are attributable to the varied parameter alone. Generation uses thirty sampling steps and a guidance scale of 7.5 throughout. Five measures are reported. Structural similarity [42] and mean absolute error are computed between the output and the input photograph, and measure how much of the input's structure and colour survive the transformation. A histogram distance is computed between the output and the reference portraits in HSV space, and two feature distances are computed from CLIP image embeddings [6]: one against the reference portraits and one against the input photograph. The histogram distance is an L1 distance between normalised histograms; both CLIP distances are cosine distances between L2-normalised embeddings. The CLIP style measure is computed pairwise against each reference portrait and the distances averaged, rather than against a distributional summary, so it remains valid at a sample size of thirty. The two style measures aggregate differently. The histogram reference is the mean histogram over all thirty portraits, computed once; the CLIP measure compares against each portrait separately and averages the resulting distances. In both cases the reference is the set rather than any single image: a distance to one portrait would measure similarity to that character's own hair, skin and clothing colours rather than to the style shared across the set. Two further figures are reported as scale references. The reference portraits differ from each other by 0.5352 on the histogram measure and 0.1797 on the CLIP measure. These give the axes a scale: a distance below the within-set spread indicates that the output lies inside the range covered by the reference set itself.

6.2 Style Acquisition

The 450-step configuration produced output that reproduced the aliased edges and blocked shading characteristic of the target style, confirming that low-rank adaptation captures the low-frequency textural signature of the idiom from thirty examples. Grid regularity, however, was inconsistent: pixel cells varied in effective size across the image and did not align to a common lattice. Extending training to approximately 900 steps at a reduced learning rate produced a marked improvement in this respect, with visibly more uniform cell sizes and a stronger tendency toward the flat colour regions that characterise hand-authored pixel art. This suggests that grid regularity is a higher-order property of the style that requires more optimisation steps to acquire than texture alone, and that the initial configuration was undertrained rather than at capacity.

6.3 Failure of Prompt-Based Colour Control

The central negative result of this project concerns attribute control. With the adapter active, colour instructions in the text prompt are substantially ignored. Requests for specific hair colour, eye colour or garment colour produced outputs whose colours appeared to be sampled independently of the instruction. A representative early result generated a subject with blonde hair and blue eyes from a prompt explicitly specifying black hair, and produced a saturated red background from a prompt specifying a white one.



Figure 6.1: Representative failure of prompt-based attribute control. The prompt specified black hair and a white background.

A second failure accompanies the first. Attempts to strengthen colour control by lengthening the prompt or by adding weighted emphasis produced not stronger adherence but structural collapse, with output degenerating toward incoherent colour fields. Reducing prompt length restored structural coherence but not colour control. The explanation consistent with both observations is the overfitting behaviour analysed in the subject-driven setting [16]. The pixel structure of the training set is an unusually strong and unusually consistent signal; with thirty images and no regularisation set, the adapter allocates its limited capacity to fitting that signal, and the residual it learns overwrites the attention pathways through which the base model associates colour tokens with spatial regions. The trigger token consequently binds to a mixture of all training images rather than to style alone, and the caption-level disentanglement designed in Section 3.4 does not survive into the learned representation. This is the reason the prompt used in the final pipeline is minimal, and the reason colour is supplied through a separate channel rather than through text.

6.4 Structural Conditioning

Three Canny threshold pairs were compared on the preprocessed portrait. The low setting produced dense contours including texture within the hair region; the high setting retained only the outer silhouette and lost the internal facial features on which identity depends. The intermediate setting captured eye, nose, mouth and garment boundaries while suppressing hair texture, and was adopted. A sweep over the conditioning scale at values of 0.4, 0.6, 0.8 and 1.0 showed a clear optimum. At 0.4 the generated facial layout departed noticeably from the edge map, with features displaced and proportions altered. At 1.0 adherence was strict but the output acquired a rigid quality, with hair in particular rendered as flat blocks that followed the contour without the internal variation the style calls for. The value 0.8 preserved feature placement while leaving the adapter sufficient freedom to impose stylistic character, and was adopted for all subsequent experiments. A separate observation from this sweep is that eye colour is unstable across all settings, appearing in saturated hues unrelated to the input. This is consistent with the attribute-binding failure described in Section 6.3 and is not remedied by structural conditioning, which carries no colour information.

6.5 Colour Conditioning: Two Strategies Compared

Given that prompt-based colour control fails, two alternative strategies were implemented and compared under identical structural and stylistic settings. The first strategy corrects colour after generation. Every pixel of the generated image is remapped to its nearest neighbour in the thirty-two colour palette extracted during preprocessing. This restores colour correspondence reliably, since the output palette is by construction a subset of the input palette. Its defect is that the palette derived from a photograph contains numerous closely spaced tones representing skin gradation, and nearest-neighbour assignment against such a palette reintroduces smooth gradients, flattening precisely the blocked shading that the adapter had produced. The strategy therefore trades style for colour. The second strategy supplies colour before generation. The quantised image is used as the initial state of the sampling trajectory [23], so that the denoiser begins from a noised version of the correct colours rather than from pure noise. Colour information is thus available to the model throughout sampling and is integrated with the style prior rather than imposed on top of it. The comparison confirms this reasoning: output retains both approximately correct colours and the blocked shading of the adapter.



Figure 6.2: Colour conditioning strategies compared. Left, the quantised colour reference; centre, post-hoc palette remapping; right, image-to-image colour conditioning.

The second strategy is adopted as the pipeline default. The first is retained as an option for cases where exact palette conformance is required and stylistic strength can be sacrificed.

6.6 Noise Strength Ablation

The noise strength parameter governs how far the sampler may depart from the supplied colour image, and a sweep at 0.3, 0.5, 0.7 and 0.9 characterises the trade-off precisely.

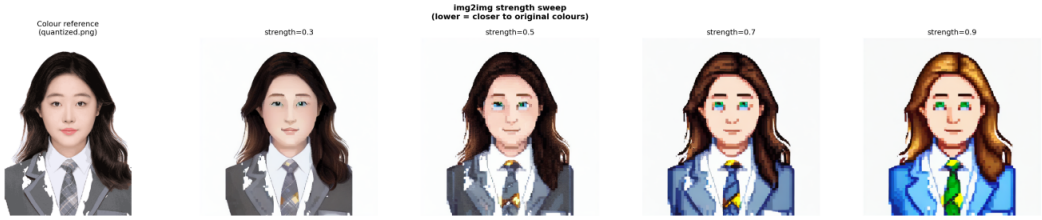


Figure 6.3: Noise strength ablation at 0.3, 0.5, 0.7 and 0.9, with the quantised colour reference shown at left.

At 0.3 the output remains close to the input photograph. Colours are faithful and identity is well preserved, but the image reads as a lightly stylised photograph rather than as pixel

art: the grid is barely visible and skin tone retains continuous gradation. At 0.9 the output is unambiguously pixel art, with strong grid structure and flat colour regions, but the colour correspondence that motivated the strategy has largely been lost, with hair shifting toward gold and garments toward saturated blue and green. The intermediate values occupy the usable region, and 0.5 was judged the best compromise, retaining recognisable hair and garment colours together with a clearly articulated pixel grid.



Figure 6.4: Representative output at the adopted operating point.

This ablation makes explicit a trade-off that is intrinsic to the task rather than to the implementation. Identity preservation and stylistic abstraction are in direct tension, because the style is defined partly by the discarding of the very detail that carries identity. The parameter does not resolve the tension; it exposes it and allows a position to be chosen deliberately.

6.7 Quantitative Comparison Against Baselines

Four configurations were evaluated. B0 is a purely algorithmic baseline: the quantised photograph is downsampled and restored by nearest-neighbour interpolation, with no learned component. B1 applies the style adapter alone, generating from text with no image input of any kind. B2 adds the control network to B1, supplying structural constraint but no colour information. The complete pipeline adds image-to-image colour conditioning to B2.

Configuration	**SSIM ↑**	**StyleDist HSV ↓**	**MAE ↓**	**StyleDist CLIP ↓**	**IdentityDist CLIP ↓**
B0 Traditional	0.7424	0.9120	8.39	0.4068	0.1647
B1 Adapter only	0.2955	1.1428	107.09	0.1645	0.5503
B2 Adapter + control	0.2371	1.3470	121.34	0.2285	0.4625
Complete pipeline	**0.6954**	**0.9091**	**17.78**	**0.2849**	**0.3281**

Table 6.1: Quantitative comparison across four configurations.

Scale reference: the thirty reference portraits differ from each other by 0.5352 on the histogram measure and 0.1797 on the CLIP measure. The HSV histogram fails to distinguish B0 from the complete pipeline. B0 scores 0.9120 and the complete pipeline 0.9091, a difference of 0.003. The measure records colour proportion and is invariant to spatial arrangement. B0 downsamples the photograph into pixel blocks using a basic algorithm, while the complete pipeline produces stylised blocks reconstructed by the generative model. The two differ substantially in structure and texture, but their overall colour composition is similar:

both are dominated by white background, skin tone and grey clothing. Because the proportions match, the histogram registers them as near-identical and the structural difference goes unmeasured. Unlike the histogram, CLIP embeddings are sensitive to texture and spatial structure as well as colour. Substituting CLIP feature distance changes both the size of the gap and the ordering of the four configurations. Distances from different metrics cannot be compared directly, since one is an L1 distance between normalised histograms and the other a cosine distance in feature space. What can be compared is each gap as a share of the range spanned by the four configurations. In HSV the four configurations span 0.44, ranging from 0.9091 to 1.3470, and the 0.003 gap between B0 and the complete pipeline is under one per cent of it. In CLIP the span is 0.24, ranging from 0.1645 to 0.4068, and the 0.12 gap is roughly half of that range. Furthermore, the ordering changes substantially. The two metrics rank the same four outputs differently at every position. The HSV histogram places B0 second and B1 third, even though B0 applies no stylisation while B1 is generated entirely by the adapter. Under CLIP, B1 rises to first and B0 falls to last, which indicates that the two metrics measure fundamentally different properties. HSV reflects colour proportion, while the defining characteristic of pixel art is spatial organisation. A distance value requires a reference scale before it can be interpreted. The thirty reference portraits differ from each other by 0.1797 on the same measure, and this figure provides a baseline. Two conclusions follow. First, B1 has a style distance of 0.1645, below the within-set spread of 0.1797. Its output therefore falls within the distribution of the reference set: in CLIP feature terms it is already a plausible Stardew Valley portrait, which indicates that the adapter has learned the style successfully. Second, B0 has an identity distance of 0.1647, also below 0.1797. Its output is therefore closer to the input photograph than the reference portraits are to each other, which quantifies the observation that B0 performs essentially no stylistic transformation.

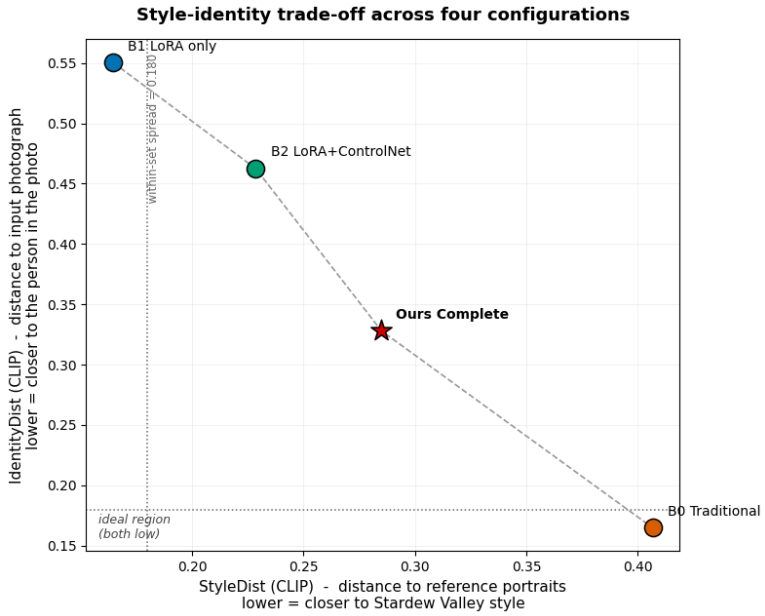


Figure 6.5: Style-identity trade-off across the four configurations. Dotted lines mark the within-set spread of the reference portraits.

The four configurations are ordered inversely on the two axes. Each additional conditioning signal moves the output one step toward better identity preservation and weaker stylisation: from B1, adding the control network gives B2, adding image-to-image conditioning gives the complete pipeline, and removing the generative component entirely gives B0. This ordering follows from the mechanism: the stronger the conditioning, the more the output is pulled back toward the input photograph, and the further it moves from the target style distribution.

The most informative feature of the plot is that the lower-left region is empty. That region would contain a configuration whose style distance and identity distance both fall below the 0.1797 baseline. No configuration falls within it. The four configurations therefore form a Pareto frontier: no configuration is better than another on both axes at once. B1 and B0 are each optimal on one axis, but neither addresses the task, which requires a portrait that is simultaneously recognisable as the subject and as a Stardew Valley character. The complete pipeline sits in the middle of the frontier and is the only configuration that is not extreme on either measure.

Chapter 7: Discussion

7.1 The Value of Decoupling

The pipeline's principal design commitment is that structure, style and colour are handled by separate mechanisms. The experimental record justifies this. Section 6.3 shows that a single mechanism, the text prompt, cannot carry all three when the style adapter is trained on thirty images: the adapter's acquisition of style actively destroys the prompt's capacity to specify colour. Section 6.4 shows that structural conditioning succeeds precisely because it is a geometric constraint expressed in a geometric medium rather than in text. Section 6.5 shows that colour is best supplied as an image for the same reason, and Section 6.7 quantifies the cost of removing either channel. The generalisable observation is that when adaptation data is scarce, the correct response is not to demand more of the learned component but to reduce what is asked of it. Each property that can be specified exactly should be specified exactly, leaving the learned component responsible only for what genuinely requires learning. In this project only stylistic character falls into that category.

7.2 What the Metrics Capture

SSIM and MAE are computed against the input photograph. This is appropriate for tasks in which the input is the intended target, such as denoising or super-resolution, but it conflicts with the aim of stylisation. A measure that rewards similarity to the input is maximised by a method that changes nothing, while stylisation necessarily involves departure from the input. The HSV histogram fails for a different reason. Its reference is the target style rather than the input. The difficulty lies in the property it measures: it records global colour proportion and is insensitive to spatial arrangement. A measure of this kind cannot separate two images with matching colour composition but different spatial organisation. CLIP feature distance succeeds because its embeddings encode texture and spatial structure alongside colour. It therefore registers the structural differences that the first two measures miss. A general point follows. An evaluation measure must be sensitive to the property that defines the target domain. For pixel art that property is spatial organisation, so any measure based on colour statistics is insufficient to evaluate it, however correctly implemented. Before adopting a quantitative criterion for style transfer, the first question is which dimension defines the target style, rather than which general-purpose similarity measure is available.

7.3 Limitations

Three limitations are apparent in the output. Eye colour remains unreliable across all configurations tested. The training portraits render eyes at a resolution of a few pixels in saturated hues, and the adapter appears to have learned this as a property of the style rather than as an attribute of the subject; the colour conditioning signal is too weak at that spatial scale to override it. Second, the proportion of the frame occupied by the subject differs systematically from the training convention. The reference portraits place the head across a large share of the canvas in a manner closer to caricature than to photographic framing, whereas the pipeline output inherits the framing of the input photograph because structural conditioning faithfully reproduces it. Matching the convention would require a reframing step in preprocessing rather than any change to the model. Third, the segmentation stage produces

intermittent errors at the boundary between dark hair and dark clothing, removing portions of the shoulder and collar. These regions become white background and are subsequently reproduced as such in the output. A salient-object segmentation network [39], or one of the more recent portrait-matting models, would be a direct substitution.

7.4 Threats to Validity

The evaluation now includes a measure of stylistic proximity, but that measure is itself a proxy. CLIP embeddings are trained predominantly on natural photographs and digital illustrations rather than on the highly abstracted domain of pixel art. While CLIP feature distance serves as a general proxy for perceptual similarity, its validity has not been calibrated against human judgement for this specific aesthetic. Because no formal user study was conducted, the configuration rankings derived from this measure must remain provisional. A second threat concerns generality. All reported results derive from portraits of a single subject in similar framing. Whether the identified operating parameters transfer across pose, lighting and demographic variation is untested, and the one non-frontal photograph examined during development produced a markedly poorer edge map, which suggests that the pipeline's usable domain may be narrower than the test set indicates. A third threat is that the qualitative judgements throughout Section 6 were made by the author alone. A formal user study, of the kind used to validate earlier pixelisation work [25], would be required before the operating points reported here could be described as generally preferred rather than as preferred by one observer.

Chapter 8: Conclusion and Future Work

Relying on a single generative mechanism is not viable under severe data constraints. Attribute control through the prompt fails once the adapter is active: colour instructions are ignored, and lengthening the prompt produces structural collapse rather than stronger adherence. The explanation consistent with both observations is that the style signal occupies the limited capacity available to the adapter, overwriting the pathways through which the base model binds colour terms to image regions. Decoupling structure, style and colour into separate mechanisms is therefore not an architectural preference but the response this failure makes necessary. The decoupled evaluation reveals a result that runs against expectation. Style acquisition is not the difficult part. The text-only configuration falls inside the feature-space range covered by the reference set itself, which indicates that the adapter learned the target style from thirty examples. The bottleneck was never that the data was too scarce to learn the style, but that preserving subject identity through a transformation this abstract is where the difficulty lies. The conflict between style and identity is structural rather than a matter of hyperparameter tuning. The four configurations form a Pareto frontier, and the region in which both distances fall below the within-set spread is empty. Each additional conditioning signal improves identity preservation and weakens stylisation. This follows from the conditioning mechanism itself: the stronger the signal, the further the output is pulled back toward the input photograph. The limitation is therefore not specific to this implementation but inherent to any method that stylises through conditional control. Four directions follow. First, a user study to validate CLIP feature distance against human judgement in this domain, since it is used here as a proxy for perceptual similarity without such validation. Second, a dataset-size ablation at fifteen, thirty and sixty images, testing whether colour binding recovers with additional training data and so converting the interpretation in Section 6.3 into a tested hypothesis. Third, replacing the general-purpose segmentation stage with a dedicated portrait-matting network to remove the boundary artefacts described in Section 7.3. Fourth, a reframing step in preprocessing to align subject scale with the compositional convention of the training set. A broader question remains open. The pipeline described here reproduces the surface characteristics of its target style but not its authorial logic: a human pixel artist decides which details to discard, and that decision is a compositional judgement rather than a filtering operation. Whether that judgement can be learned from data, or whether it must remain a human contribution, is beyond the scope of this project but is the question its results most directly raise.

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